Environmental Particle Extension

Sun/Wind Dual-Kind Carriers, Root Chaos, and Leaf Generation

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Abstract

We extend the VSEPR-SIM bead-based framework with a dual environmental particle regime consisting of radiative *sun particles* and advective *wind particles*. Sun particles deposit directional energy into exposed beads through projected-area and shading-dependent transfer kernels. Wind particles contribute momentum loading, drying, and transport bias. Root development is modulated by a bounded stochastic response factor $\Gamma_i \in [0.98, 1.8]$ combined with piecewise polynomial moisture response and logic-gated growth modifiers. Leaf generation follows a Heaviside step-function initiation gate with post-threshold continuous expansion. The system composes additively with the existing wind-particle perturbation infrastructure.

Contents

1 Environmental Particle State Vector

Definition 1.1 (Environmental particle)**.** *Each environmental particle is a lightweight carrier state:*

$$P_k = (\text{id}, \text{ kind}, \mathbf{r}_k, \mathbf{v}_k, E_k, I_k, \omega_k, \chi_k, \tau_k) \tag{1}$$

where:

- $\mathbf{r}_k$*: position*
- $\mathbf{v}_k$*: direction / transport velocity*
- E_k*: energy content*
- I_k*: local intensity weight*
- ω_k*: oscillation or temporal modulation term*
- χ_k*: stochastic or chaos control parameter*
- τ_k*: lifetime / persistence*

The kind field partitions particles into two classes:

$$\text{kind} \in \{\,\textsc{Sun}, \textsc{Wind}\,\} \tag{2}$$

Sun particles are not literal photons. They are radiative packets or directional energy carriers. Wind particles are not fluid elements. They are momentum plus transport disturbance packets.

2 Sun Particle Deposition

For a bead i, the energy deposited by sun particle k:

$$\Delta E_i^{\text{sun}} = \sum_{k\in\text{SUN}} A_i^{\text{proj}}\, I_k\, T_i\, S_i^{\text{photo}}\, \Phi_{\text{shade}}(i,k) \tag{3}$$

Symbol	Meaning
A_i^{proj}	Projected area toward incident radiation
T_i	Tissue transmissivity or absorptivity
S_i^{photo}	Biological photo-response coefficient
$\Phi_{\text{shade}}(i,k)$	Shadowing / occlusion factor $\in [0,1]$

For leaves, sun deposition drives:

- Growth activation
- Local expansion bias
- Surface heating
- Transpiration tendency

For roots, sun is indirect and arrives through temperature field coupling, moisture gradient response, and shoot-driven resource allocation.

3 Wind Particle Force

$$\Delta \mathbf{F}_i^{\text{wind}} = \sum_{k \in \text{WIND}} C_{d,i}\, A_i^{\text{eff}}\, \|\mathbf{v}_k - \mathbf{v}_i\|^2\, \hat{\mathbf{n}}_{ik}\, \Psi_i^{\text{flex}} \tag{4}$$

Symbol	Meaning
$C_{d,i}$	Drag-like bead coefficient
A_i^{eff}	Effective exposed area
$\hat{\mathbf{n}}_{ik}$	Loading direction (unit relative velocity)
Ψ_i^{flex}	Structural flexibility response

Wind drives not only force but also: mechanical bending, oscillatory fatigue, pollen transport, evaporation and drying, boundary-layer disturbance, seed release probability.

The drying rate from wind:

$$\dot{D}_i = \kappa_{\text{dry}}\, \|\mathbf{v}_k - \mathbf{v}_i\|\, A_i^{\text{eff}} \tag{5}$$

4 Root Chaos Factor

4.1 Bounded Stochastic Multiplier

The root exploration multiplier:

$$\Gamma_i = \text{clamp}\big(\alpha_0 + \alpha_1 M_i + \alpha_2 \nabla W_i + \alpha_3 \rho_i^{\text{soil}} + \alpha_4 \chi_i,\ 0.98,\ 1.8\big) \tag{6}$$

Symbol	Meaning	Default
α_0	Base offset	1.0
α_1	Moisture weight	0.3
α_2	Water gradient weight	0.15
α_3	Soil density penalty	-0.2
α_4	Stochastic amplitude	0.1

This gives:

- Near-deterministic roots in stable media
- More chaotic probing in poor or heterogeneous zones
- Occasional lateral overreach and branching competition

4.2 Piecewise Polynomial Response

$$R_{\text{root}}(x) = \begin{cases} a_1 x^2 + b_1 x + c_1 & x < x_{\text{dry}} \\ a_2 x^3 + b_2 x^2 + c_2 x + d_2 & x_{\text{dry}} \le x < x_{\text{opt}} \\ a_3 x^2 + b_3 x + c_3 & x \ge x_{\text{opt}} \end{cases} \tag{7}$$

The piecewise polynomial is wrapped in logic:

$$g_i^{\text{root}} = \text{clamp}\big(R_{\text{root}}(M_i) \cdot L_{\text{gates}}(s_i) \cdot \Gamma_i,\ 0.05,\ 2.25\big) \tag{8}$$

where L_{gates} applies multiplicative penalties:

$$\begin{aligned} \text{if } D_i > D_{\max}: &\quad \times 0.55 \\ \text{if } \rho_i^c > \rho_{\max}: &\quad \times 0.72 \\ \text{if } \nabla N_i > N_{\text{trig}}: &\quad \times 1.18 \\ \text{if } S_i^{\text{sun}} < S_{\min}: &\quad \times 0.91 \end{aligned} \tag{9}$$

5 Leaf Generation Gate

5.1 Heaviside Initiation

$$H_{\text{leaf}} = H\big(\beta_1 E_{\text{local}} + \beta_2 S_{\text{light}} + \beta_3 T_{\text{hydr}} - \beta_4 \Sigma_{\text{damage}} - \Theta_{\text{leaf}}\big) \tag{10}$$

Below threshold: no leaf initiation. Above threshold: leaf primordium appears.

5.2 Continuous Expansion

After initiation, growth becomes continuous:

$$\frac{dA_{\text{leaf}}}{dt} = H_{\text{leaf}} \cdot G_{\text{leaf}}(E, L, W, \Sigma) \tag{11}$$

where G_{leaf} depends on energy availability, light, water, and structural stress.

6 Energy Decomposition

The total environmental energy:

$$U_{\text{env}} = U_{\text{wind}} + U_{\text{sun}} + U_{\text{dry}} + U_{\text{photo}} + U_{\text{stress}} \tag{12}$$

Each term is explicitly tracked and inspectable, consistent with the anti-black-box design philosophy.

7 Particle Advection and Lifetime

Particles advance by:

$$\mathbf{r}_k(t + \Delta t) = \mathbf{r}_k(t) + \mathbf{v}_k \, \Delta t \tag{13}$$

$$\tau_k(t + \Delta t) = \tau_k(t) - \Delta t \tag{14}$$

With optional oscillation modulation:

$$I_k \leftarrow I_k \cdot \big(1 + 0.1 \sin(\omega_k \, \Delta t)\big) \tag{15}$$

A particle is removed when $\tau_k \leq 0$.

8 Integration with Existing Wind Particle System

The existing `WindParticle` (from `wind_particle.hpp`) operates on the atomistic `State` directly. The environmental particle extension operates at the bead level. The two compose:

1. `WindParticle::apply(state)` adds force to atomistic state
2. `interact_env_particle(bead, p)` computes plant-bead response
3. Both contribute to U_{ext} additively

The same codebase serves both metallic/turbine and plant workflows.